KUMARAGURU COLLEGE OF TECHNOLOGY

Department of Artificial Intelligence and Data Science

DATA SCIENCE AND VISUALIZATION (24ADI204)

WEEK 2 REPORT

Know Your Data

Youth Smoking and Drug Consumption Analysis

DATA-VIZ HACK-A-BIT 2026–2027

Team 16

Team Member Name	Roll Number
Anireetha M	25BAD009

1. Week 2 Contribution

Identified features suitable for scaling, encoding and PCA.

2. Work Carried Out

During Week 2, my individual work focused on understanding the UCI Drug Consumption dataset using the project notebook. I examined the dataset structure, data types, missing values, duplicate records and unique categories. Based on this inspection, I identified the numerical and categorical features that may be considered for scaling, encoding and PCA in the later preprocessing stages.

3. Features Identified

- **Numerical features for scaling – Identified the personality score columns Nscore, Escore, Oscore, Ascore, Cscore, Impulsive and SS as numerical features. These can be considered for scaling after checking their ranges and distributions.**
- **Features for PCA – The seven personality score columns (Nscore, Escore, Oscore, Ascore, Cscore, Impulsive and SS) were identified as possible inputs for PCA because they represent numerical personality-related attributes. Scaling should be considered before applying PCA.**
- **Categorical features for encoding – Identified the drug-consumption columns Alcohol, Amphet, Amyl, Benzos, Caff, Cannabis, Choc, Coke, Crack, Ecstasy, Heroin, Ketamine, Legalh, LSD, Meth, Mushrooms, Nicotine, Semer and VSA. These columns contain consumption categories such as CL0 to CL6 and may require suitable encoding.**
- **Demographic features – Age, Gender, Education, Country and Ethnicity are stored as numerical codes in the dataset. Their categorical or ordinal meaning should be considered before deciding whether to scale or encode them. The ID column is an identifier and is not a useful feature for PCA.**
- **Dataset inspection – The notebook showed 1,885 rows and 32 columns, with 19 object columns, 12 float64 columns and 1 int64 column. Missing values were found in Age, Gender, Education and Nscore (113 in each column), while no duplicate rows were found.**

4. Outcome of My Contribution

This inspection helped me identify the seven personality-score variables as candidate features for scaling and PCA, and the 19 drug-consumption variables as categorical features that may require encoding. I also recognized that the demographic codes need to be interpreted carefully before preprocessing. These findings provide a clear starting point for the next stages of feature preparation. This report describes feature identification and does not claim that I personally implemented scaling, encoding or PCA during Week 2.

5. Reference

Project notebook: Drug_Consumption_Know_Your_Dataipynb (1).ipynb.